Microbial diversity in Mediterranean sponges as revealed by metataxonomic analysis

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Abstract

In the present work, we firstly aimed at exploring the biodiversity of marine sponges from four different areas of the Mediterranean: Faro Lake in Sicily and "Porto Paone", "Secca delle fumose", "Punta San Pancrazio" in the Gulf of Naples. Eight sponge species were collected from these sites and identified by morphological analysis and amplification of several conserved molecular markers (18S and 28S RNA ribosomal genes, mitochondrial cytochrome oxidase subunit 1 and internal transcribed spacer). In order to analyse the bacterial diversity of symbiotic communities among these different sampling sites, we also performed a metataxonomic analysis through an Illumina MiSeq platform, identifying more than 1500 bacterial taxa. Amplicon Sequence Variants (ASVs) analysis revealed a great variability of the host-specifc microbial communities.

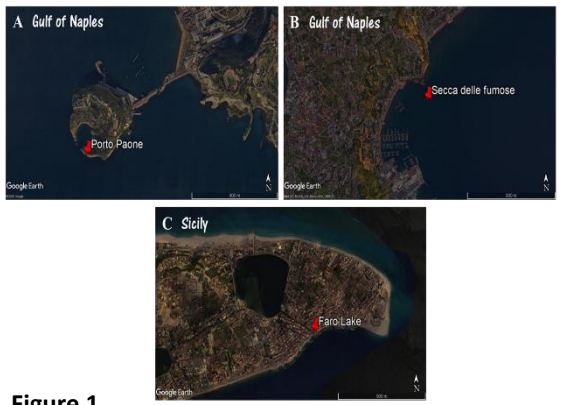


Figure 1

Conclusions

Our data highlight the occurrence of dominant and locally enriched microbes in the Mediterranean, together with the biotechnological potential of these sponges and their associated bacteria as sources of bioactive natural compounds.

References

Ruocco N., Esposito R., Zagami G., Bertolino M., De Matteo S., Sonnessa M., Andreani F., Crispi S., Zupo V., Costantini M. (2021) Marine microbiota as revealed by metagenomics analysis on Mediterranean sponges. Scientific Reports 11:21151

Results

Morphological and molecular analyses

- All studied sponges belong to the class Demospongiae, commonly live in the Mediterranean Sea, with the only exception of *Oceanapia cf. perforata* (Sarà, 1960), which represents a rare species distributed in the western Mediterranean.
- BLAST similarity search corresponded with the morphological identification, defining that the sponges are: *Tethya aurantium*, *Axinella damicornis*, *Oceanapia* cf. *perforata*, *Sarcotragus spinosulus* and *Agelas oroides*.

Metataxonomic analysis (Ruocco, Esposito et al., 2021)

- Heat-map, comparing the abundance of the most representative bacterial classes, showed that sponges from Faro Lake have a species-specific associated community, while the sponges collected in Gulf of Naples showed a site-specific bacterial composition (Figure 2)
- Krona Plot represents the most abundant phyla for each sponge under analysis (Figure 3)

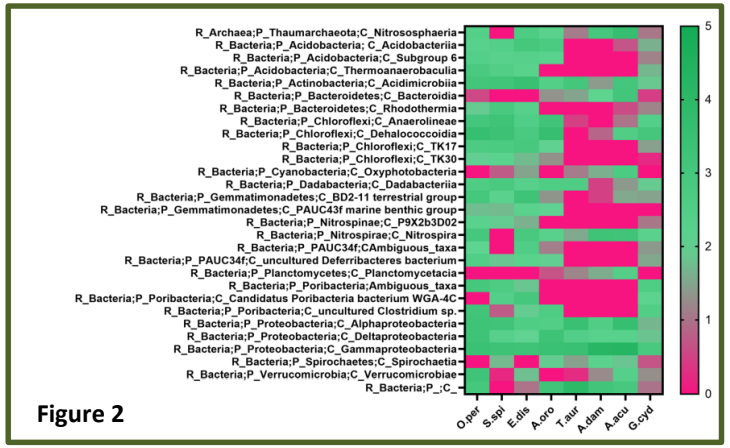
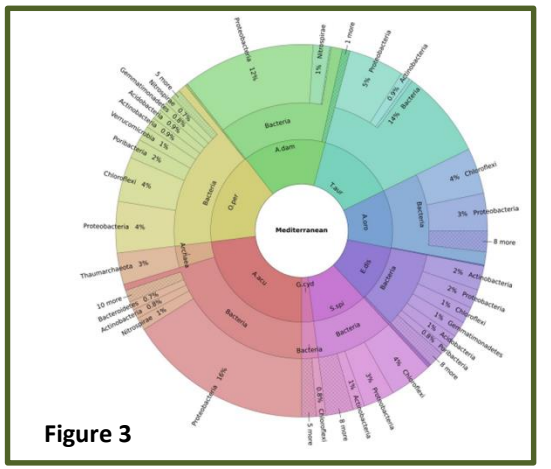


Figure 2


Mediterranean
Bacteria
Archaea
Proteobacteria 12%
Chloroflexi 4%
Proteobacteria 4%
Thaumarchaeota 3%
Chloroflexi 4%
Proteobacteria 16%
Proteobacteria 5%
Chloroflexi 4%
Proteobacteria 3%
Actinobacteria 2%
Proteobacteria 2%
Chloroflexi 1%
Gemmatimonadetes 1%
Acidobacteria 1%
Chloroflexi 4%
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